Exercise 3 4.11.2025

Problem 7

For measuring RF power amplifiers, a measurement setup based on fig. 1 is used. The following table gives the characteristics of the setup.

Max. power of the signal generator	$P_{sig,max} = 20$ mW
Gain of the preamplifier	$G_{pre} = 400$
Max. input power of the preamplifier	$P_{1,max} = 16$ mW
Coupling loss	$a_C = 20$ dB
Loss of the attenuator	$a_A = 22$ dB
Max. input power of the powermeter	$P_{4,max} = 45$ mW
Gain of the device under test (DUT)	$G_{DUT} = 10 \ldots 40$

a) Convert all powers from the table into dBm and all gains into dB.

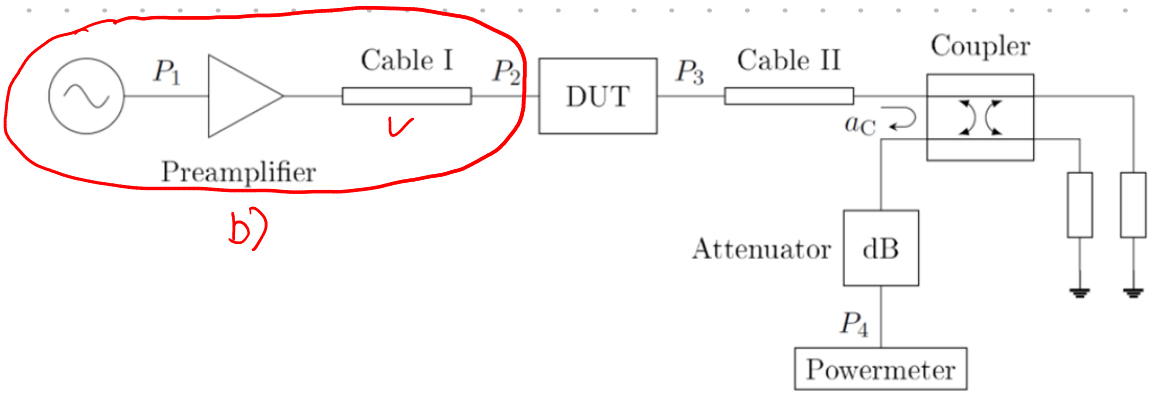
$P_{sig,max}|_{dBm} = 10 \cdot \log \frac{20\,mW}{1\,mW} = 10 \cdot \log \left(\frac{2 \cdot 10\,mW}{1\,mW}\right) = 13\,dBm$

$G_{pre}|_{dB} = 10 \cdot \log 400 = 10 \log 2 \cdot 2 \cdot 10 \cdot 10 = 26\,dB$

$P_{1,max}|_{dBm} = 10 \cdot \log \frac{16\,mW}{1\,mW} = 12\,dBm$

$P_{4,max}|_{dBm} = 10 \cdot \log \frac{45\,mW}{1\,mW} = 16.5\,dBm$

$G_{DUT}|_{dB} = 10 \ldots 16\,dB$



b)

Figure 1 Schematic of the measurement setup

b) $P_1|_{dBm} = P_{sig} = 10 \cdot \log \frac{5\,\mu W}{1\,mW} = -23\,dBm$

$P_2 = 10 \cdot \log \frac{1.8\,mW}{1\,mW} = 2.6\,dBm$

$P_2 = P_1 + G_{pre} - a_I$

$a_I = P_1 + G_{pre} - P_2 = -23\,dBm + 26\,dB - 2.6\,dBm = 0.4\,dB$

c)

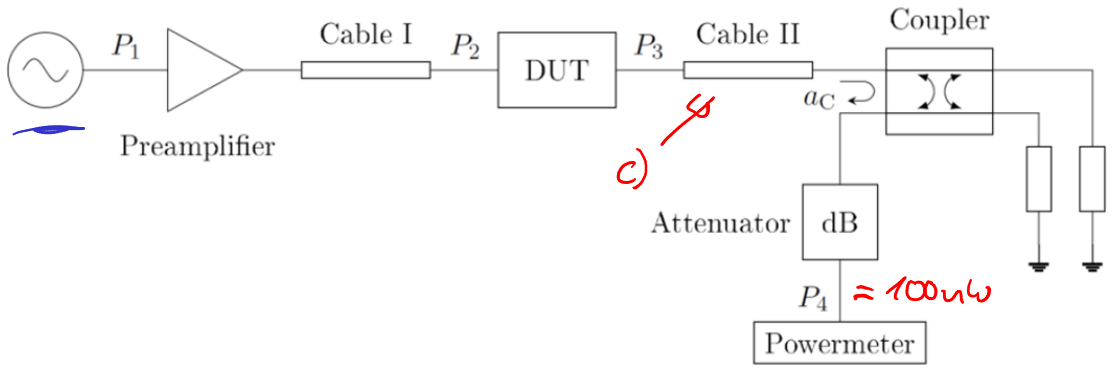


Figure 1 Schematic of the measurement setup

$P_4 = 10 \cdot \log \frac{100\mu W}{1mW} = -40\,dBm$

$P_4 = P_3 - a_{II} - a_C - a_A$ $\quad P_2 = P_3$ e.g. no device under test

$a_{II} = P_3 - a_C - a_A - P_4 \approx 2.6\,dBm - 20dB - 22dB - (-40dBm) = 0.6dB$

d) $P_2 \overset{!}{=} 1W \quad P_2 = 10 \cdot \log \frac{1W}{1mW} = 30\,dBm$

$P_1 = P_2 + a_I - G_{Pre} = 30dBm + 0.4dB - 26dB = 4.4\,dBm$

$P_1 = 10^{\frac{4.4\,dBm}{10}} \cdot 1mW = 2.75\,mW$

e) $P_4 \approx 0.8\,dBm$

$P_4 = P_2 + G_{DUT} - a_{II} - a_C - a_A$

$\Rightarrow G_{DUT} = P_4 - P_2 + a_{II} + a_C + a_A = 0.8dBm - 30dBm + 0.6dB + 20dB + 22dB$

$= 13.4\,dB$

$G_{DUT}|_{lin} = 10^{13.4dB/10} = 21.9$ //

Problem 5

An electromagnetic wave propagates in a waveguide, e.g. an optical fiber, with a loss of 0.4 dB/km. Find the distances that the wave has to travel until the power decays to 1/2 (1/10, 1/100) of its initial value.

Definitions: $P(z) = P_0 e^{-2\alpha \cdot z}$ $[\alpha] = 1/m$ amplitude absorption coefficient

Power: $P[dBm] = 10 \cdot \log \frac{P[W]}{1mW} \Leftrightarrow P[mW] = 10^{P[dBm]/10} \cdot 1mW$

Attenuation: $a[dB] = 10 \cdot \log \frac{P_0[W]}{P_{(z-z_0)}[W]} = \frac{1mW}{1mW} =$

$$10 \cdot \log \frac{P_0[W]}{1mW} - 10 \log \frac{P(z-z_0)[W]}{1mW} = P_0[dBm] - P_{z=z_0}[dBm]$$

Attenuation per length: $\alpha_{dB}\left[\frac{dB}{m}\right] = \frac{a[dB]}{z_0[m]}$

$P_{(z)} = P_0 e^{-2\alpha z} \Rightarrow 2\alpha z = \ln \frac{P_0}{P_{(z)}}$ $\ln x = \frac{\log x}{\log e}$

$$2\alpha z = \frac{10 \cdot \log \frac{P_0}{P(z)}}{10 \cdot \log e} = \frac{a[dB]}{10 \cdot \log e}$$

$$\alpha = 1/2 \cdot \frac{a[dB]}{z} \cdot \frac{1}{10 \log e} = 1/2\, \alpha_{dB}\left[\frac{dB}{m}\right] \cdot \frac{1}{\log e}$$

$$\alpha_{dB}\left[\frac{dB}{m}\right] = 2 \cdot \alpha\left[\frac{1}{m}\right] \cdot 10 \cdot \log e \approx a\left[\frac{1}{m}\right] \cdot 8.686\ dB$$

$$\alpha_{dB} = \frac{\alpha}{z} = \frac{10 \cdot \log \frac{P_0}{P(z)}}{z} \Rightarrow z = \frac{10 \cdot \log \frac{P_0}{P(z)}}{\alpha\ dB}$$

$\alpha_{dB} = 0.4\ dB/km$

$\frac{P(z)}{P_0} = \frac{1}{2} \Rightarrow \frac{P_0}{P(z)} = 2$ $z_1 = \frac{10 \cdot \log 2}{0.4\ dB/km} = 7.52\ km$

$\frac{P(z)}{P_0} = \frac{1}{10} \Rightarrow 25\ km$

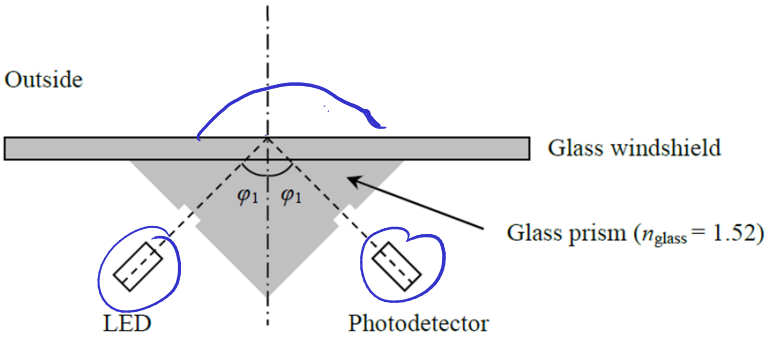
$\frac{P(z)}{P_0} = \frac{1}{100} \Rightarrow 50\ km$

Problem 6

A simplified graph of a rain sensor is illustrated in the figure below. The rain sensor is fixed on the glass windshield of a car ($n_{glass} = 1.52$). The LED and the photodetector are positioned so that $\varphi_1 = 45°$.

1.47 = n_2

1.48 = n_1



a) $n_1 \cdot \sin\varphi_1 = n_2 \cdot \sin\varphi_2$

critical angle describes the incident angle φ_1, where φ_2 becomes 90°

$n_1 \cdot \sin\varphi_1 = n_2 \cdot \sin 90° =$ $\sin\varphi_1 = \sin\varphi_c = \frac{n_2}{n_1}$ with $n_2 < n_1$

$\varphi_c = \sin^{-1}\frac{n_{air}}{n_{glass}} = \sin^{-1}\frac{1}{1.52} = 41.14°$

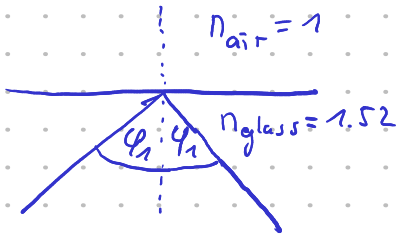
$\varphi_1 = 45° > \varphi_c = 41.14°$ => total internal reflection (TIR)

$r_e = r_h = 1$

perpendicular polarisation refl. coefficient

parallel polarisation refl. coeff.

$R_e = |r_e|^2 = 1 = R_h = |r_h|^2$

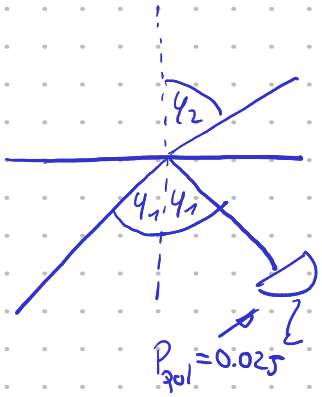


$\sin\varphi_c = \frac{n_w}{n_{glass}}$ => $\varphi_c = \sin^{-1}\frac{1.33}{1.52} = 61.04°$

$\varphi_1 = 45° < \varphi_c = 61.04°$ => reflection and refraction

$\varphi_2 = \sin^{-1}\left(\frac{n_{glass}}{n_w} \cdot \sin\varphi_1\right) = \sin^{-1}\left(\frac{1.52}{1.33} \cdot \sin 45°\right)$

$\varphi_2 = 53.91°$



Power reflection coefficient:

$R_e = |r_e|^2 = \left(\frac{\sin(\varphi_2 - \varphi_1)}{\sin(\varphi_2 + \varphi_1)}\right)^2 = \left(\frac{\sin(53.91° - 45°)}{\sin(53.91° + 45°)}\right)^2 = 0.025$ /

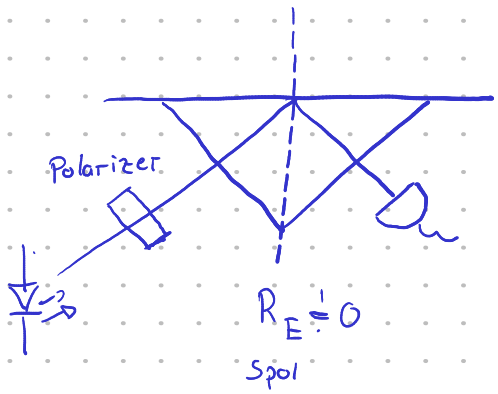
P_{pol}

$R_H = |r_h|^2 = \frac{\cos(\varphi_2 + \varphi_1) \cdot \sin(\varphi_2 - \varphi_1)}{\cos(\varphi_2 - \varphi_1) \cdot \sin(\varphi_2 + \varphi_1)} = 0.0006$

S_{pol}

difference maybe useful ...

c)



Reflected power needs to be minimized, what is on the windscreen.

d) further improvement $R_{H} \doteq 0$ $\gamma_1 = \gamma_B$ set γ_1 to be the Brewster angle.

$$\gamma_B = \tan^{-1}\left(\frac{n_w}{n_{glass}}\right) = \tan^{-1}\left(\frac{1.33}{1.52}\right) = 41.19°$$